

AWS offers a broad spectrum of compute offerings that give you the flexibility to choose the right tool for the right job. The three main categories of compute are *virtual machines*, *containers*, and *serverless*. There is no one-size-fits-all service because it depends on your needs. The key is to understand what each option has to offer in order to build a more appropriate cloud architecture for your use case. In this unit, you learn about containers and how to run them on AWS. Containers can host a variety of different workloads, including web applications, lift and shift migrations, distributed applications, and streamlining of development, test, and production environments.

While containers are often referred to as a new technology, the idea started in the 1970s with certain Linux kernels having the ability to separate their processes through isolation. At the time, this was configured manually, making operations complex. With the evolution of the open source software community, containers evolved. Today, containers are used as a solution to problems of traditional compute, including the issue of getting software to run reliably when it moves from one compute environment to another. A container is a standardized unit that packages up your code and all of its dependencies. This package is designed to run reliably on any platform, because the container creates its own independent environment. This makes it easy to carry workloads from one place to another, such as from development to production or from on-premises to the cloud.

When you hear the word *container*, you may associate it with Docker. Docker is a popular container runtime that simplifies the management of the entire operating system stack needed for container isolation, including networking and storage. Docker makes it easy to create, package, deploy, and run containers.

The diagram illustrates the architectural differences between Virtual Machines (VMs) and Containers. It is divided into two main sections: Virtual machines (VMs) and Containers.

Virtual machines (VMs): This section shows three separate VMs stacked vertically. Each VM consists of an Application layer (Application 1, Application 2, Application 3) and a Guest OS layer (Guest OS). These VMs are managed by a Virtualization platform, which sits on top of the Host operating system, which in turn runs on the Server hardware.

Containers: This section shows three separate Containers stacked vertically. Each Container consists of an Application layer (Application 1, Application 2, Application 3) and a Container OS layer (Container OS). These containers are managed by a Containerization platform, which sits on top of the Operating system, which in turn runs on the Server hardware.

Virtual machines (VMs)			Containers		
VM	VM	VM	Container	Container	Container
Application 1	Application 2	Application 3	Application 1	Application 2	Application 3
Guest OS	Guest OS	Guest OS	Container OS	Container OS	Container OS
Virtualization platform			Containerization platform		
Host operating system			Operating system		
Server			Server		

ORCHESTRATE CONTAINERS

- How to place your containers on your instances.
- What happens if your container fails.
- What happens if your instance fails.
- How to monitor deployments of your containers

MANAGE CONTAINERS WITH AMAZON ELASTIC CONTAINER SERVICE (AMAZON ECS)

The diagram illustrates the relationship between ECS and EC2. On the left, the ECS logo is shown. A line connects it to a large rounded rectangle labeled 'Cluster' at the bottom. Inside the 'Cluster' rectangle, there are two smaller rounded rectangles, each labeled 'EC2 container instance' at the top. Each 'EC2 container instance' rectangle is divided into four quadrants, with two of them containing icons representing container instances.

The diagram illustrates the architecture of an Amazon ECS cluster. On the left, the ECS logo is shown. A line connects it to an EC2 container instance icon. This instance is part of a cluster, represented by a large rounded rectangle. Inside this rectangle, there are three EC2 container instance icons and an ECS Agent icon at the bottom. The entire group is labeled 'Cluster' at the bottom.

Here is a simple task definition that you can use for your corporate director application. In this example, the runs on the Nginx web server.

USE KUBERNETES WITH AMAZON ELASTIC KUBERNETES SERVICE (AMAZON EKS)

- An EC2 instance with the ECS Agent installed and configured is called a container instance. In Amazon EKS, it is called a worker node.
- An ECS Container is called a task. In the Amazon EKS ecosystem, it is called a pod.
- While Amazon ECS runs on AWS native technology, Amazon EKS runs on top of Kubernetes.

Resources:

- [External Site: AWS: Containers on AWS](#) 
- [External Site: Docker: What Is a Container?](#) 
- [External Site: AWS: Amazon Elastic Container Service](#) 
- [External Site: Github: Amazon ECS Agent](#) 
- [External Site: AWS: Amazon ECS container instances](#) 
- [Coursera Course: Building Containerized Applications on AWS](#) 

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